DEMO SESSION 3 · MILESTONE 3

MILESTONE 3 Requirements

Present your insights, your preprocessing pipeline, and a dataset that is genuinely ready for modeling.

WELCOME BACK

Quick Recap of Module 3

1. Matplotlib charts: line, bar, pie, histogram, scatter, box plot
2. Seaborn charts and correlation heatmaps
3. Descriptive statistics and probability basics
4. Feature selection, engineering, and encoding
5. Standardization, normalization, and train/test splits
6. The ML workflow, overfitting, and underfitting

TODAY'S GOAL

There is no new coding today. Present your EDA and preprocessing work, and prove your dataset is truly ready for modeling.

What Is Milestone 3?

DEFINITION

Milestone 3 is your third checkpoint, where you prove your data is explored, prepared, and genuinely ready for a model.

This is the last stop before Module 4, where you'll actually start training Machine Learning models.

What Happens in This Demo Session

◎ **DEFINITION**

Each team presents their EDA and preparation work, and receives feedback on the spot.

1. Your team walks through your key insights and preprocessing decisions
2. The instructor and your peers question your feature engineering and leak-free checks
3. You are graded on rigor and clarity, not on having already built a model

PART 2 · WHAT YOU SUBMIT

Your Dataset & Visualizations

1. Your clean dataset, carried forward from Milestone 2

2. The key visualizations that reveal what's happening in your data

PART 2 · WHAT YOU SUBMIT

Your Analysis

1. The business insights your exploration revealed

2. Your correlation analysis, and what it does and doesn't prove

PART 2 · WHAT YOU SUBMIT

Your Preparation Work

1. Your feature engineering, and the reasoning behind each new feature
2. Your full data preprocessing pipeline, from encoding through splitting
3. Your final dataset, genuinely ready for modeling

PART 2 · WHAT YOU SUBMIT

Your Deliverables

1. An EDA notebook showing your exploration and charts
2. A data preprocessing notebook showing your full pipeline
3. A presentation summarizing insights and decisions

PART 2 · WHAT YOU SUBMIT

What Makes a Dataset 'Ready for Modeling'

◎ **DEFINITION**

A dataset is ready for modeling when every column is numeric, scaled sensibly, and split without leakage.

1. Every category column has been encoded into numbers
2. Every numeric column has been scaled or normalized appropriately
3. The train and test sets are split with zero overlap, and no leakage between them

Good vs Risky Preprocessing Decisions

Signal	What It Looks Like
Scaling order	Good: split first, then fit the scaler on training data only. Risky: scaled before splitting
Feature engineering	Good: each new feature has a stated reason. Risky: features added without explanation
Encoding	Good: one-hot or label encoding chosen deliberately. Risky: encoding applied without checking for leakage
Final dataset	Good: fully numeric, no leakage, documented. Risky: still contains raw text or ID columns

Before-and-After Snapshot

Metric	Before → After
Columns	12 → 9 (irrelevant columns dropped)
Category columns	3 text columns → 7 encoded numeric columns
Feature scale	Raw units → standardized (mean 0, std 1)
Rows in train / test	0 / 0 → 634 / 159 (no overlap)

INSTRUCTOR TIP

A Note From Your Instructor



Confirm your final dataset is genuinely leak-free and your train/test split is finalized.

This is the last checkpoint before modeling begins — issues caught here are far cheaper than issues caught in Module 4.

Milestone 3 Rubric

Criteria	What We Look For
Insight quality	Clear, specific business insights — not just charts without conclusions
Correlation analysis	Correct interpretation, with no confusion between correlation and causation
Feature engineering	Each new feature has a stated, sensible reason for existing
Preprocessing pipeline	Leak-free, in the correct order, and clearly documented
Final dataset readiness	Fully numeric, properly scaled, and split without overlap

Sample Insight Statement

Weak Version	Strong Version
"Income and target are correlated."	"Customers earning above $60k are 2.3x more likely to convert, suggesting income-tiered offers."
"We removed some outliers."	"We capped 11 income values above the IQR upper bound to prevent them from skewing scaling."

JUST IN CASE

Common Milestone 3 Mistakes

Mistake	What Happens
Scaling before splitting	The most common leakage mistake at this milestone — always split first
Presenting charts without conclusions	A chart alone isn't an insight — the takeaway needs to be stated
Overclaiming causation	A strong correlation is described as if it proves cause, which it doesn't
Skipping a final readiness check	The dataset is assumed ready without actually verifying shapes and types

QUICK RECAP

What a Strong Milestone 3 Looks Like

1. Clear, specific business insights
2. Correlation interpreted correctly, not as causation
3. Feature engineering with a stated reason
4. A leak-free, well-ordered preprocessing pipeline
5. A final dataset that's fully numeric and properly split
6. A team that can defend every decision under questions

Good Luck